

Andres Bonilla

Software Engineer

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SUMMARY

Software Engineer with 3+ years of experience building full-stack web applications, REST APIs, automated testing infrastructure, and internal tooling across Python, C#/.NET, TypeScript/React, and SQL. Comfortable owning features end-to-end — from database design through CI/CD deployment and monitoring. Co-author on a peer-reviewed publication in computational biophysics simulation.

TECHNICAL SKILLS

Languages: Python, C#, .NET, JavaScript, TypeScript, SQL, C++, R, Java

Frontend & Backend: React, REST API, HTML, Entity Framework, SQLAlchemy, Gunicorn, Nginx

Infrastructure: Linux, Azure DevOps, CI/CD, Git, GitHub, logging, monitoring

Databases: SQL Server, schema design, query optimization, database performance

Tooling: GitHub Copilot, Claude, Microsoft Copilot 365, prompt engineering

WORK EXPERIENCE

Baylor Genetics | Senior Software Engineer

June 2025 – Present

- Served as the primary technical point of contact for a high-visibility genomic reporting system redesign — drove architectural direction for team's implementation, coordinated requirements across multiple upstream and downstream teams, and ensured cross-functional alignment throughout delivery.
- Had significant influence over the SQL database schema and C#/.NET REST API design, shaping shared data contracts and integration standards that enabled parallel development across dependent teams without creating hard blockers.
- Mentored teammates and resolved cross-functional technical blockers day-to-day, contributing to consistent code quality and on-time delivery under active stakeholder oversight.

Baylor Genetics | Software Engineer

June 2024 – June 2025

- Owned a critical internal database and web tool end-to-end — resolved systemic data quality issues that virtually eliminated support tickets from downstream teams and restored cross-team confidence in the platform.
- Built a Python/SQLAlchemy backend for an internal data analysis web app with query logging and performance tracking, designed to replace a manual genomic data workflow for an internal operations team.
- Delivered full-stack features across multiple tools using C#/.NET REST APIs and TypeScript/React, deployed new internal applications to Linux via Azure DevOps CI/CD pipelines with Nginx and Gunicorn.

Baylor Genetics | Analytical Software QA Engineer

August 2023 – June 2024

- Designed and built a modular Python application for automated regression testing of REST APIs and databases, automating 60% of the previously manual test suite and enabling reliable monthly QA cycles.
- Created Azure DevOps CI/CD pipelines and diagnostic dashboards to surface test results and failure data to engineering and operations teams.
- Partnered with internal and external teams to onboard new services into QA infrastructure as the organization's systems evolved.

University of Michigan – Michigan Medicine | Software Developer

September 2021 – May 2023

- Extended the Cytosim C++ physics simulation engine with novel microtubule breakage mechanics, contributing to a peer-reviewed publication in the Journal of Cell Biology (2025).
- Automated large-scale research data processing using R and Slurm cluster computing, substantially reducing experiment turnaround time.

EDUCATION

University of Michigan, Ann Arbor

August 2019 – April 2023

B.S. with Honors: Molecular Biology · Minor in Computer Science

Grand Prize Winner – EECS 183 Computer Science Showcase (Fall 2019)